

JEE MAIN REVISION MASTER: GRAVITATION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 14 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Newton's Law of Gravitation & Acceleration Due to Gravity (g)	9 PYQs	Gravitational Field
Moderate	Orbital Velocity, Escape Speed & Kepler's Laws	5 PYQs	Orbits & Satellites

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Newton's Law of Gravitation & Acceleration Due to Gravity (g)	$g = \frac{GM}{R^2}$, $g_h \approx g\left(1 - \frac{2h}{R}\right)$, $g_d = g\left(1 - \frac{d}{R}\right)$, $g_{\text{pole}} - g_{\text{equator}} = \omega^2 R$
Orbital Velocity, Escape Speed & Kepler's Laws	$v_e = \sqrt{2gR} = \sqrt{2} v_o$, $\left(\frac{T_1}{T_2}\right)^2 = \left(\frac{r_1}{r_2}\right)^3$, $E_{\text{total}} = -K = \frac{U}{2} = -\frac{GMm}{2r}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Newton's Law of Gravitation & Acceleration Due to Gravity (g) (High Yield — 9 PYQs)

Core Principle: Gravitational acceleration at surface: $g = (GM / R^2)$. Variation with height: $g_h = g(1 - 2h/R)$ (for $h \ll R$) or $g_h = g(R^2 / (R+h)^2)$. Variation with depth: $g_d = g(1 - d/R)$ (strictly linear). At center of earth ($d=R$), $g=0$. Effect of rotation: $g' = g - \omega^2 R \cos^2\lambda$.

■■ **Caution / JEE Pitfall:** Height variation decreases g twice as fast as depth: at depth $d = 2h$, the decrease in g matches the decrease at height h .

■ **Speed Trick:** Weight is maximum at the poles and minimum at the equator due to centrifugal relief from Earth's axial rotation.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Suppose the acceleration due to gravity at the earth's surface is gm/s^2 and at the surface of moon it is g^m/s^2 " [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Imagine a new planet having the same density as that of the earth, but it is two times bigger than the earth in size. If the acceleration du..." [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "What is the value of acceleration due to gravity at a height equal to half the radius of the Earth, from its surface ?..." [Ans: N/A]

3.2 Orbital Velocity, Escape Speed & Kepler's Laws (Moderate Yield — 5 PYQs)

Core Principle: Escape velocity from surface $v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR} \approx 11.2 \text{ km/s}$. Orbital velocity for close orbit $v_o = \sqrt{gR} \approx 7.9 \text{ km/s}$. Key ratio: $v_e = \sqrt{2} v_o$. Kepler's third law: $T^2 \propto r^3$. Geostationary satellite: period 24 hours, height $\approx 36,000 \text{ km}$, orbits west to east.

■ **Caution / JEE Pitfall:** Escape velocity is completely INDEPENDENT of the mass of the projectile and the angle of projection!

■ **Speed Trick:** Total energy of an orbiting satellite is negative: $E = -K = U/2$. If orbital speed is increased by 41.4% (factor of $\sqrt{2}$), the satellite escapes Earth's gravity.

Representative Source PYQs:

- **[2025 Shift] Q#3 (MCQ):** "If r_p, v_p, L_p and r_a, v_a, L_a are radii, velocities and angular momenta of a planet at perihelion and aphelion of its elliptic..." [Ans: N/A]
- **[2025 Shift] Q#4 (MCQ):** "The total energy of a satellite in a circular orbit at a distance $(R + \Delta R)$ from the centre of the Earth varies as ΔR [R is the radius of the ...]" [Ans: N/A]
- **[2023 Shift] Q#2 (MCQ):** "When a planet revolves around the Sun, in general, for the planet..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Suppose the acceleration due to gravity at the earth's surface is $g \text{ m/s}^2$ and at the surface of moon it is $g' \text{ m/s}^2$. An $M \text{ kg}$ passenger goes from the earth to the moon in a spaceship moving with a constant velocity (Neglect all other objects in the sky). Which curve best represents the weight (net gravitational force) as a function of time?

- (A)
- (B)
- (C)
- (D)

Correct Answer: None

Solution Analysis: Let the mass of the Earth be M_e , the mass of the Moon be M_m , and the distance between their centres be d . Suppose the spaceship is at a distance r from the centre of the Earth. Then its distance from the centre of the Moon is $(d-r)$. The Earth pulls the passenger towards the Earth, and the Moon pulls the passenger towards the Moon. These two pulls are in opposite directions, so the net gravitational force on the passenger of mass M is:

$$F_{\text{net}} = \frac{GM_e M}{r^2} - \frac{GM_m M}{(d-r)^2}$$
 The spaceship moves with constant velocity. So the distance r from the Earth in

Benchmark Problem 2: KCET 2025 (Q#3)

If r_p, v_p, L_p and r_a, v_a, L_a are radii, velocities and angular momenta of a planet at perihelion and aphelion of its elliptical orbit around the Sun respectively, then

- (A) $r_p > r_a, v_p > v_a, L_a > L_p$
- (B) $r_p < r_a, v_p > v_a, L_a = L_p$
- (C) $r_p > r_a, v_p < v_a, L_a = L_p$
- (D) $r_p < r_a, v_p < v_a, L_a < L_p$

Correct Answer: None

Solution Analysis: Angular momentum is conserved in an elliptical orbit, meaning: $L_p = L_a$ where L_p is the angular momentum at perihelion and L_a is the angular momentum at aphelion. The perihelion is the point in the orbit where the planet is closest to the Sun. Therefore, the radius at perihelion (r_p) is less than the radius at aphelion (r_a):

$r_p < r_a$ Because angular momentum is conserved, we have: $mv_p r_p = mv_a r_a$ From this, it follows that the velocity at perihelion (v_p) is greater than the velocity at aphelion (v_a): $v_p > v_a$